

Wide-Angle Multislice for Electron Microscopy via the Wave Propagation Method

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Abstract

Multislice simulation is the standard method for modeling electron beam-sample interactions in transmission electron microscopy (TEM). The traditional multislice algorithm is calculated using a split-step Fourier method that employs a paraxial (Fresnel) propagator, limiting its accuracy to small scattering angles (typically $< 7^\circ$) and ignoring the change of the electron wavelength inside the sample potential. These limitations can be significant in scenarios involving high-angle scattering or thick, high-Z samples. In this work, we present a comparison of three practical multislice propagators of increasing physical complexity: Fresnel multislice, angular spectrum multislice, and the Wave Propagation Method (WPM) from the light optics community, benchmarked against a forward ODE solver for the Klein-Gordon equation. We write Klein-Gordon, in its Helmholtz form, adapt the WPM to electron microscopy via a differentiable refractive-index binning strategy, and benchmark the methods against high-angle reference solutions in two settings: the Rother-Scheerschmidt Au [100] beam-amplitude verification and convergent-beam diffraction in Au [110].

1 Introduction

Multislice simulation has been the standard tool for modelling electron propagation through thin specimens for several decades, and in its usual Fresnel form it underpins essentially all quantitative image and diffraction calculations in transmission electron microscopy (TEM). It is fast, conceptually simple, and accurate enough that comparisons with experiment are typically limited by other factors, such as the quality of the atomic potential or of the microscope characterisation, rather than by the propagator itself.

Recent developments in the field have begun to push this picture. Aberration correctors routinely produce probes with semi-convergence angles of tens of milliradians; fast pixelated detectors record full diffraction patterns at every probe position, opening up 4D-STEM and electron ptychography as quantitative imaging modalities; and there is growing interest in using multislice simulation as a differentiable forward model inside reconstruction pipelines. In these settings the simulated exit wave or diffraction pattern is compared directly with the experiment, and small systematic errors at high scattering angles can no longer be safely absorbed into a fitting parameter. The accuracy of the propagator at non-paraxial angles therefore matters in a way that it did not for traditional image-mode TEM.

The limits of the Fresnel propagator in this regime are well known. The kernel is obtained by replacing the exact free-space dispersion relation $k_z = \sqrt{k_0^2 - k_\perp^2}$ by its leading paraxial term, and as a result it loses accuracy when a significant fraction of the wave's energy sits at transverse wavevectors that are no longer small compared to k_0 . Several alternatives have been investigated. One can solve the full second-order wave equation directly, but this is too expensive to use as a routine forward model. Wave Propagation Method (WPM) schemes from the light-optics community keep the split-step structure of multislice but additionally allow the local wavelength to change inside the specimen. In between sits a much simpler option already common in X-ray ptychography: leave the multislice transmission step untouched and swap only the Fresnel kernel for the angular spectrum kernel, which retains the exact vacuum dispersion relation while preserving the same FFT-based per-slice cost.

The aim of this work is to ask, in the specific context of electron microscopy, how much of the wide-angle error is removed by that minimal change, and how much of the remainder is recovered by the more elaborate WPM. To do this we benchmark Fresnel multislice, angular spectrum multislice,

and WPM against a direct ODE solution of the full second-order Klein–Gordon equation in two test cases: the Au [100] beam-amplitude verification of Rother and Scheerschmidt [1], which we extend to thicker specimens, and a convergent-probe CBED study at semi-angles up to 100 mrad. The Klein–Gordon ODE makes no split-step or paraxial approximation and serves as a common ground truth, so the differences between the three split-step schemes can be attributed cleanly to their respective propagation kernels.

The physics of an electron traversing an electrostatic potential $V(\mathbf{r})$ without spin effects can be calculated by the Time Independent Klein Gordon wave equation, which can be written with an effective refractive index

$$\nabla^2 \psi + k_0^2 n^2(\mathbf{r}) \psi = 0, \quad (1)$$

where $k_0 = \sqrt{E^2 - m_0^2 c^4} / \hbar c$ is the vacuum wavenumber and

$$n(\mathbf{r}) = \sqrt{\frac{(E - V(\mathbf{r}))^2 - m_0^2 c^4}{E^2 - m_0^2 c^4}} \quad (2)$$

is the effective electron refractive index set by the local electrostatic potential. The index is unity in vacuum and departs from it only where the potential is nonzero, encoding the full relativistic energy–momentum relation. Expanding the potential-dependent term in $k_0^2 n^2$ gives

$$k_0^2 n^2(\mathbf{r}) = k_0^2 - \frac{2EV(\mathbf{r})}{\hbar^2 c^2} + \frac{V^2(\mathbf{r})}{\hbar^2 c^2}. \quad (3)$$

For $|V| \ll E$ the quadratic term is small, and using $E = \gamma m_0 c^2$ equation (1) reduces to the relativistically corrected Schrödinger equation

$$\nabla^2 \psi + \left(k_0^2 - \frac{2\gamma m_0}{\hbar^2} V(\mathbf{r}) \right) \psi = 0, \quad (4)$$

equivalent to the linearised refractive index $n(\mathbf{r}) \approx 1 - (\gamma m_0 / \hbar^2 k_0^2) V(\mathbf{r})$. Typically the potential is sufficiently weak that the linearised index is accurate to better than 1% for all V encountered in TEM, so the Schrödinger equation is a good approximation to the Klein-Gordon equation.

The traditional method of choice to solving either (1) or (4) is to ignore backscattered effects, and slice the potential into thin slices with a free space propagation inbetween - a.k.a the multislice method. In multislice, the specimen is sliced into thin layers of thickness Δz , each layer applying the transmission factor $\exp(i\sigma V_{\text{proj}})$ with $V_{\text{proj}}(x, y) = \int_{\text{slice}} V dz$ and σ the relativistic interaction constant; between slices the wave is propagated through empty space. Once the thin-slice transmission step is fixed, the only remaining modelling choice is the free-space propagator. Setting $V = 0$ in equation (4) and splitting the Laplacian as $\nabla^2 = \partial_z^2 + \nabla_\perp^2$ leaves

$$(\nabla_\perp^2 + \partial_z^2 + k_0^2) \psi = 0, \quad (5)$$

with z the beam direction and (x, y) the transverse coordinates. The exact free-space update of equation (5) is the angular spectrum propagator, which applies the spherical dispersion relation $k_z = \sqrt{k_0^2 - k_\perp^2}$ to each transverse Fourier component. Expanding this square root to leading order,

$$\sqrt{k_0^2 - k_\perp^2} = k_0 - \frac{k_\perp^2}{2k_0} + \mathcal{O}\left(\frac{k_\perp^4}{k_0^3}\right),$$

and absorbing the carrier phase $e^{ik_0 \Delta z}$ yields the familiar Fresnel propagator. The two free-space propagators therefore differ only in how the longitudinal dispersion is treated:

$$\underbrace{\exp\left(i\sqrt{k_0^2 - k_\perp^2} \Delta z\right)}_{\text{angular spectrum: exact dispersion}} \quad \text{vs.} \quad \underbrace{\exp\left(-i\frac{k_\perp^2}{2k_0} \Delta z\right)}_{\text{Fresnel: parabolic dispersion}}. \quad (6)$$

Combining the same thin-slice transmission step with the angular spectrum kernel yields angular spectrum multislice; combining it with the Fresnel kernel and writing the result as a continuous equation yields the parabolic (paraxial) Schrödinger equation

$$\frac{\partial \psi}{\partial z} = \frac{i}{2k_0} \nabla_\perp^2 \psi + i\sigma V(x, y, z) \psi, \quad (7)$$

which standard paraxial multislice solves. The Bloch-wave formulation traditionally applies the same parabolic linearisation about the incident beam direction, and as a result conventional Bloch-wave and Fresnel multislice are two numerical realisations of the same one-way paraxial approximation.

The cost of this approximation has been examined repeatedly in the electron microscopy literature. Chen and Van Dyck derived an accurate multislice scheme directly from the Schrödinger equation without making the high-energy approximation: factoring out the carrier wave but retaining the second derivative in z , the envelope equation factorises into forward- and backward-propagating branches whose forward slice propagator contains an operator-valued square root that preserves the spherical dispersion relation. Because the transverse Laplacian and the projected potential do not commute, that operator cannot be split into the usual Fourier-space propagator and real-space transmission product, and the square root must instead be approximated as an operator or matrix function—at substantially higher cost than conventional multislice. Wacker and Schröder give a clear derivation of this construction and benchmark it against conventional Fresnel multislice and direct Runge–Kutta/Lobatto integration of the non-approximated Schrödinger equation, showing that dropping the second z -derivative introduces substantial errors at low voltages, large thicknesses, and high scattering angles. Rother and Scheerschmidt approached this problem from a relativistic perspective: their comparison of Dirac, Klein–Gordon, and relativistically corrected Schrödinger calculations confirms that the dominant high-angle error is the parabolic propagation step of the fresnel multislice method itself.

The light and X-ray optics communities have addressed the same square-root operator from a different perspective, and several of their solutions have not been made explicit in electron microscopy. In X-ray ptychography it is now standard to retain the angular spectrum kernel in free space while keeping the specimen interaction in split-step form, which recovers high-angle propagation almost for free. Padé approximant methods replace the square root by a rational function that remains accurate to larger angles than the Fresnel expansion while preserving the Fourier-domain split-step structure. Wave Propagation Method (WPM)-type schemes go further still and admit a local, potential-dependent longitudinal wavelength inside the medium, restoring the change of wavelength that the paraxial reduction discards. These techniques span a useful accuracy–cost trade-off that has been partially explored but not systematically benchmarked for the electron-microscopy regime.

2 Numerical methods

We compare four models of increasing physical complexity—standard Fresnel multislice, angular spectrum multislice, the Wave Propagation Method, and a direct forward-ODE solver for the full second-order Klein–Gordon equation (1). The three approximate solvers are benchmarked against the Klein–Gordon ODE reference in two high-angle settings: the Au [100] beam-amplitude benchmark of Rother and Scheerschmidt, which tracks a 135 mrad beam through a thick gold crystal, and a convergent-beam electron diffraction (CBED) geometry in Au [110], where we test each method as a function of probe convergence semi-angle up to 100 mrad. In every case the specimen is divided into thin slices of thickness Δz perpendicular to the optic axis and the wavefunction is advanced slice-by-slice on a transverse $N_x \times N_y$ grid; the methods differ only in how the slice propagator is constructed. The forward-ODE solver makes no split-step or one-way approximation and serves as the numerical ground truth throughout.

2.1 Forward Klein-Gordon ODE ground truth

The most accurate scheme retains the full second-order Klein-Gordon equation (1) without making a one-way approximation. Separating the Laplacian as $\nabla^2 = \partial_z^2 + \nabla_\perp^2$, equation (1) reads

$$\partial_z^2 \psi = -(\nabla_\perp^2 + k_0^2 n^2(x, y, z)) \psi, \quad (8)$$

which we recast as a first-order system in the state (ψ, ϕ) with $\phi \equiv \partial_z \psi$,

$$\frac{d}{dz} \begin{pmatrix} \psi \\ \phi \end{pmatrix} = \begin{pmatrix} \phi \\ -(\nabla_\perp^2 + k_0^2 n^2(x, y, z)) \psi \end{pmatrix}, \quad (9)$$

and integrate along z with an adaptive high-order Runge–Kutta scheme (8th-order Dormand–Prince). The right-hand side is evaluated in $\mathcal{O}(N \log N)$ per stage by handling each term in its diagonal basis:

$\nabla_{\perp}^2 \psi$ is computed via two FFTs (multiplication by $-k_{\perp}^2$ in Fourier space) while $k_0^2 n^2(x, y, z) \psi$ is a pointwise real-space multiplication. The potential is treated as piecewise constant over each slice of thickness Δz and the step-size controller is clipped at the slice boundaries so the solver never steps across a discontinuity in $n^2(z)$. The wavefunction is initialised with the input probe $\psi(z=0)$ and the longitudinal derivative

$$\phi(z=0) = \mathcal{F}^{-1}[i\sqrt{k_0^2 - k_{\perp}^2} \mathcal{F}\{\psi(z=0)\}], \quad (10)$$

which enforces purely forward-propagating waves in the vacuum entrance plane. This can be viewed as a real-space, arbitrary-probe generalisation of the forward-scattering ODE benchmark used by Rother and Scheerschmidt [1], without the periodic-crystal and beam-basis restrictions of that formulation. Because no split-step or paraxial factorisation is applied, the scheme reproduces the full dispersion relation $k_z = \sqrt{k_0^2 n^2 - k_{\perp}^2}$ and correctly couples the transverse Laplacian to the spatially varying potential within each slice. Its cost is dominated by the FFT pairs of the right-hand side evaluations; the adaptive controller refines the step inside the specimen, so the ODE solver is typically several times more expensive than the split-step schemes at matched accuracy, but no more so. We use it as the ground-truth reference throughout this work.

2.2 Wave Propagation Method

The WPM, introduced by Brenner [2] and developed into a systematic wave-optical scheme for thick inhomogeneous elements by Schmidt et al. [3], starts from the exact slice propagator associated with the corresponding one-way wave equation,

$$\psi(z + \Delta z) = \exp\left(i\Delta z \sqrt{k_0^2 n^2(x, y) + \nabla_{\perp}^2}\right) \psi(z).$$

For a homogeneous slice, n is constant and therefore commutes with $\nabla_{\perp}^2$, so the square-root operator is diagonal in Fourier space. Each transverse Fourier component then propagates independently with longitudinal wavevector $k_z = \sqrt{k_0^2 n^2 - k_{\perp}^2}$, and the one-slice update is obtained by multiplying the angular spectrum by $\exp(ik_z \Delta z)$. Schmidt et al. derive in full in the appendix of their paper how this homogeneous-medium operator can be extended to a thick inhomogeneous element by partitioning the transverse plane into subdomains of nearly constant refractive index and applying the corresponding homogeneous-slice propagator locally. The resulting approximation can be written in the continuum limit as a mixed real/Fourier-space propagator,

$$\psi(x, y, z + \Delta z) = \frac{1}{(2\pi)^2} \iint \tilde{\psi}(k_x, k_y, z) e^{ik_z(x, y; k_{\perp}) \Delta z} e^{i(k_x x + k_y y)} dk_x dk_y, \quad (11)$$

with the spatially dependent longitudinal wavevector

$$k_z(x, y; k_{\perp}) = \sqrt{k_0^2 n^2(x, y) - k_{\perp}^2}. \quad (12)$$

In Schmidt's derivation, this is precisely how one goes beyond the thin-element approximation: propagation and refraction are treated simultaneously through the local dispersion relation instead of being factorised into a pure transmission function and a separate vacuum propagator. Unlike the standard split-step, the WPM kernel depends *simultaneously* on real-space coordinates (through $n(x, y)$) and Fourier-space coordinates (through $k_{\perp}$), so the propagator cannot be evaluated by a single FFT pair. The direct implementation costs $\mathcal{O}(N^2)$ per slice, which is prohibitive at typical TEM grid sizes. Following [3, 4] we sample $n(x, y)$ by a small set of reference values $\{n_m\}_{m=1}^{N_{\text{bin}}}$ so that the WPM step decomposes into a bank of homogeneous-slice angular spectrum propagations followed by a spatially weighted recombination,

$$\psi(x, y, z + \Delta z) \approx \sum_{m=1}^{N_{\text{bin}}} w_m(x, y) \mathcal{F}^{-1}\left[e^{i\Delta z \sqrt{k_0^2 n_m^2 - k_{\perp}^2}} \mathcal{F}\{\psi(x, y, z)\}\right], \quad (13)$$

where $w_m(x, y)$ interpolates the local refractive index between adjacent bins. This reduces the per-slice cost to $\mathcal{O}(N_{\text{bin}} N \log N)$. We use the differentiable binning scheme described in Appendix A, which selects $\{n_m\}$ on a quadratic grid in $[n_{\text{min}}, n_{\text{max}}]$ and uses a smoothstep weight w_m so that the propagator remains differentiable with respect to the potential. When $n(x, y)$ is constant across the transverse plane a single bin suffices and the WPM reduces exactly to the angular spectrum method described next.

2.3 Angular spectrum multislice

The angular spectrum method (ASM) replaces the paraxial Fresnel kernel of equation (6) with the exact free-space dispersion relation while retaining the split-step structure of standard multislice. The homogeneous-slice propagator in Fourier space is

$$\hat{P}_{\text{ASM}}(k_{\perp}; \Delta z) = \exp(i\Delta z \sqrt{k_0^2 - k_{\perp}^2}), \quad (14)$$

and the one-slice update is

$$\psi(x, y, z + \Delta z) \approx e^{i\sigma V_{\text{proj}}(x, y)} \mathcal{F}^{-1}[\hat{P}_{\text{ASM}}(k_{\perp}; \Delta z) \mathcal{F}\{\psi(x, y, z)\}]. \quad (15)$$

The transmission factor is identical to standard multislice. Replacing only the Fresnel kernel costs nothing: the FFT pair is unchanged and the per-slice cost remains $\mathcal{O}(N \log N)$. The relationship to Fresnel multislice is made transparent by expanding the square-root operator,

$$\sqrt{k_0^2 + \nabla_{\perp}^2} = k_0 + \frac{\nabla_{\perp}^2}{2k_0} - \frac{\nabla_{\perp}^4}{8k_0^3} + \dots,$$

where truncating at the second term recovers the Fresnel kernel $\exp(-ik_{\perp}^2 \Delta z / (2k_0))$, whereas ASM retains the operator to all orders. ASM is the minimal correction to standard multislice that captures the correct free-space high-angle propagation while preserving its split-step structure and FFT efficiency. It differs from the WPM only in that, inside the potential, the wavelength is held at its vacuum value $\lambda_0 = 2\pi/k_0$ rather than tracking the local $\lambda(x, y) = 2\pi/(k_0 n(x, y))$.

2.4 Fresnel multislice

For completeness we recall the standard split-step Fresnel multislice scheme derived in the Introduction (equation (6)): the slice propagator combines a transmission factor $\exp(i\sigma V_{\text{proj}})$ with the paraxial Fresnel kernel $\exp(-ik_{\perp}^2 \Delta z / (2k_0))$, obtained by truncating the square-root operator at second order in $k_{\perp}/k_0$. It is the cheapest of these schemes—one FFT pair per slice—and the standard tool of quantitative TEM simulation. Its accuracy is bounded by the paraxial assumption $k_{\perp}^2 \ll k_0^2$.

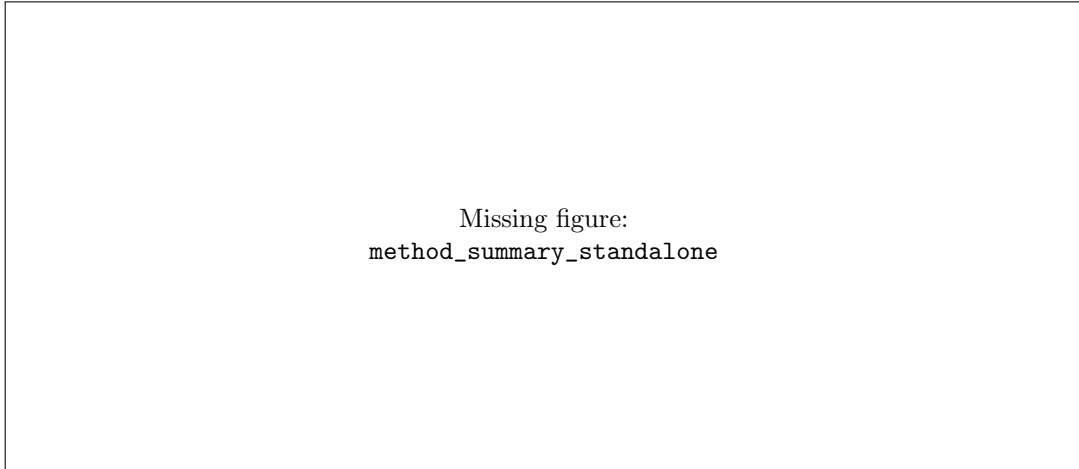


Figure 1: Relationship between the Klein-Gordon/Helmholtz wave equation, the KG ODE reference, and the three multislice propagators compared in this work. The KG ODE solves the second-order equation directly; WPM approximates the one-way square-root operator using local refractive-index propagators; angular spectrum multislice is the homogeneous-index propagation limit; and Fresnel multislice follows from the small-angle expansion of the angular-spectrum kernel.

3 Results

We present two numerical experiments that isolate the high-angle behaviour most relevant to this work. Both compare Fresnel multislice (FMS), angular spectrum multislice (ASMS), and the Wave Propaga-

tion Method (WPM) against a converged ODE solution of the full second-order Klein-Gordon equation. The first reproduces the Au [100] beam-amplitude benchmark of Rother and Scheerschmidt [1]; the second tests a convergent Au [110] probe over a wide range of semi-angles.

3.1 Verification against Rother & Scheerschmidt (2009)

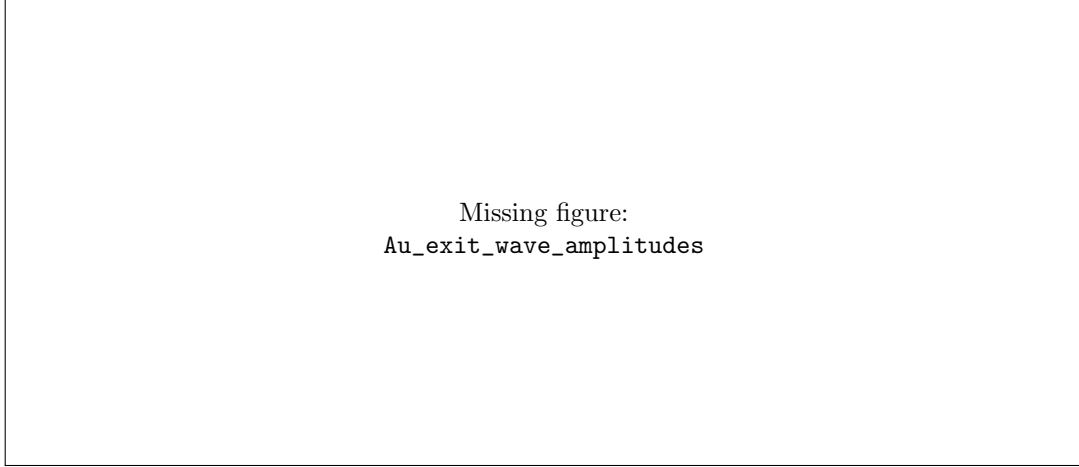


Figure 2: Exit-wave amplitude comparison for the Au [100] verification case, showing the corresponding amplitudes of the fourier transform of the exit wave produced by each propagation model and by the Klein-Gordon ODE reference.

To verify our implementation, and assess which propagation methods retain accuracy at high angles, we recreated the simulation presented in figure 3 of [1], which propagates a plane wave input through a thick gold [100] crystal at 300 keV. The ground truth is the converged ODE solution of the full second-order Klein-Gordon equation. In this example, periodic boundary conditions are implicit, due to each method using a Fourier transform solution, and as a result the crystal is infinitely wide. Since we wished to explore the accuracy of each method in thicker objects, we extended the simulation as presented by Rother and Scheerschmidt from 25 unit cells (10nm thickness) to 100 unit cells (40nm thickness) It is immediately obvious in 2, which displays the amplitude of the fourier transform of the exit wave after propagation through 100 unit cells of the gold crystal, that the Fresnel Propagator method is less accurate than the three other methods. The angular spectrum multislice, WPM, and KG ODE solutions all closely track each other, with the the WPM solution appearing to track more accurately the amplitude of the pixels in the ground truth solution compared to the angular spectrum method. However, as noted by Rother and Scheerschmidt, the images can hardly be compared quantitatively by eye: as a result, we have repeated their experiment in figure-3 by plotting the normalised amplitude of the fourier transform of the exit wave after each slice for the two selected pixels - an on axis [0, 0] pixel in the centre of the image, and a high-angle [0, 28] pixel at 135mrad from 0 to 25, and from 0 to 100 unit cells.

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Au_pixel_amplitudes

Figure 3: Pixel-amplitude comparison for the Au [100] verification case. The central and high-angle pixels are compared across Fresnel multislice, angular spectrum multislice, the Wave Propagation Method, and the Klein–Gordon ODE reference.

For the central pixel [0, 0], all methods agree closely up to 10nm thickness or 25 unit cells. We extended this simulation out to 100 unit cells or 40 nm thickness, and found that past 10nm the error between the fresnel propagator and all other methods begins to increase even for the on-axis [0,0] pixel. Past 15nm of thickness, there has been enough propagation through the crystal that higher angle diffracted beams can make their way back towards the central pixel and modify it's value. The fresnel propagator is not capable of tracking these beams, and thus it's accuracy suffers as a result. For the high-angle pixel [0, 28] at 135 mrad, the Fresnel multislice solution diverges from the ground truth after approximately 10 unit cells or 4nm, consistent again with the findings of [1] who attributed the discrepancy to the linearization of the square-root operator. Interestingly, the angular spectrum solution which requires only a small implementation change to the standard Fresnel Multislice method, tracks the ground truth much more closely than the Fresnel solution, and is comparable to the WPM solution which is significantly more expensive to implement and compute. It is only at a very large thickness of 35 nm in the off axis pixel that it becomes clear the angular spectrum is starting to deviate from the ground truth. The WPM solution tracks the ground truth more closely than the angular spectrum method, but the difference is small and only becomes apparent at very large thicknesses. At least in this case, the extra computational effort of the wave propagation method does not appear to provide a significant accuracy improvement over the angular spectrum method.

3.2 Convergent-probe Au CBED

Next, we tested each method's accuracy as a function of probe convergence semi-angle for a convergent-beam electron diffraction (CBED) geometry, on a 3 times enlarged version of the Au [100] unit cell in x and y compared to the unit cell used in the previous section up to a thickness of 100 nm. In order to ensure accurate tracking of the wave for a large convergence angle, we increased the sampling to 256 x 256 over the enlarged unit cell. The probe is focused on the entrance surface of the specimen, so the initial wavefunction in amplitude is an airy function without any aberrations. The input wave is then propagated through the specimen using each method and compared to the KG ODE ground truth. The CBED diffraction pattern is calculated at the exit surface by taking the fourier transform of the exit wave and compared across methods.

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 cbed_combined_amplitude_log_error_maps

Figure 4: CBED exit-wave amplitude comparison in log scale for the Au [100] convergence-angle study. The maps show the final diffraction patterns produced by Fresnel multislice, angular spectrum multislice, the Wave Propagation Method, and the Klein–Gordon ODE reference.

A series of convergent probes including the following semi-angles - 25, 50, 100 mrad were propagated through using 128 slices per unit cell for each method. Figure 4 again confirms that both the wave propagation method and the angular spectrum method are capable of tracking the high-angle scattering more accurately than the Fresnel multislice method. Visually the final CBED pattern of the Fresnel Propagator solution is noticeably worse than the WPM or Angular Spectrum methods. The WPM for all three convergence angles at the end of the propagation routine shows the smallest error in the CBED amplitude - staying on average at 3.5% mean error compared to the ground truth ODE solution. The WPM solution tracks the ground truth more closely than the angular spectrum method, but again difference is small and only becomes apparent at very large thicknesses. Again, and also in this more extreme case, the extra computational effort of the wave propagation method does not appear to provide a significant accuracy improvement over the Angular Spectrum method.

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Figure 5: Relative CBED error as a function of specimen thickness for the Au [100] convergence-angle study. The error is reported against the Klein–Gordon ODE reference at the end of each unit cell for the tested probe semi-angles.

In Figure 5 we visualise the root mean square (RMSE) error of the CBED intensity against the KG ODE reference as a function of object thickness, where we again see that after a thickness of approximately 10nm in all cases, there is an order of magnitude error difference in the RMSE of the Fresnel multislice method compared to the WPM and Angular Spectrum methods. The angular spectrum method does become less accurate as the thickness increases, but still remains closer to the accuracy of the WPM compared to the Fresnel Propagation Method. The WPM method tracks the

ground truth more closely than the angular spectrum method and at least up to these thicknesses, maintains a consistent relative error. Repeatedly, there does not appear to be a compelling reason to use the WPM method over the angular spectrum method, given the small accuracy improvement and the significantly higher computational cost of the WPM method. Especially given the fact that the angular spectrum method is a simple modification to the standard Fresnel multislice method, and can be implemented with no additional computational cost, it appears to be the best choice for high-angle scattering simulations in TEM.

4 Sampling Requirements

When replacing the Fresnel kernel with the angular spectrum kernel, the main concern is a question of sampling. Are the typical existing FFT grid and slice thicknesses still fine enough to represent the propagator? For Fresnel multislice this is usually straightforward, because the kernel $\exp(-ik_{\perp}^2 \Delta z / 2k_0)$ is smooth for all $k_{\perp}$. The angular spectrum kernel $\exp(i\Delta z \sqrt{k_0^2 - k_{\perp}^2})$ is slightly more delicate: its phase changes rapidly as $k_{\perp}$ approaches k_0 , and for $k_{\perp} > k_0$ the modes become evanescent. As a result, the usual phase-sampling condition gives a slightly tighter slice-thickness bound for ASM than for Fresnel. This factor comes from comparing how quickly the propagator phase changes across transverse-frequency space. For one propagation step, the Fresnel and ASM phases are

$$\phi_F(k_{\perp}) = -\frac{\Delta z}{2k_0} k_{\perp}^2, \quad \phi_{ASM}(k_{\perp}) = \Delta z \sqrt{k_0^2 - k_{\perp}^2}.$$

Their phase gradients are therefore

$$\left| \frac{d\phi_F}{dk_{\perp}} \right| = \Delta z \frac{k_{\perp}}{k_0}, \quad \left| \frac{d\phi_{ASM}}{dk_{\perp}} \right| = \Delta z \frac{k_{\perp}}{\sqrt{k_0^2 - k_{\perp}^2}}.$$

So the ASM phase varies faster by the ratio

$$\frac{|d\phi_{ASM}/dk_{\perp}|}{|d\phi_F/dk_{\perp}|} = \frac{k_0}{\sqrt{k_0^2 - k_{\perp}^2}}.$$

Evaluating this at the largest represented transverse wavevector $k_{\perp, \max}$ gives the slightly tighter ASM slice-thickness bound, by the factor

$$\frac{k_0}{\sqrt{k_0^2 - k_{\perp, \max}^2}}.$$

In conventional TEM this extra restriction is usually negligible. For the 200 keV case discussed here, the numbers are:

$$\lambda \approx 0.025 \text{ \AA}, \quad k_0 = \frac{2\pi}{\lambda} \approx \frac{2\pi}{0.025} \approx 250 \text{ \AA}^{-1}.$$

For a 200 mrad probe, the largest transverse wavevector is

$$k_{\perp, \max} = k_0 \sin \alpha \approx 250 \sin(0.2) \approx 50 \text{ \AA}^{-1}.$$

So the relevant ratio is

$$\frac{k_{\perp, \max}}{k_0} \approx \frac{50}{250} \approx 0.2.$$

The extra ASM sampling penalty is therefore

$$\frac{k_0}{\sqrt{k_0^2 - k_{\perp, \max}^2}} = \frac{1}{\sqrt{1 - (k_{\perp, \max}/k_0)^2}} \approx \frac{1}{\sqrt{1 - 0.2^2}} \approx 1.02,$$

which means the slice-thickness bound is tightened by only about 2%.

The same conclusion holds for the transverse grid. If the real-space pixel size is $\Delta x = 0.04 \text{ \AA}$, then the Nyquist spatial frequency is

$$f_{\text{Nyq}} = \frac{1}{2\Delta x} = \frac{1}{0.08} = 12.5 \text{ \AA}^{-1},$$

so the corresponding transverse wavevector is

$$k_{\perp, \text{Nyq}} = 2\pi f_{\text{Nyq}} = \frac{\pi}{\Delta x} \approx \frac{\pi}{0.04} \approx 78.5 \text{ \AA}^{-1}.$$

This gives a Nyquist scattering angle of

$$\theta_{\text{Nyq}} = \arcsin\left(\frac{k_{\perp, \text{Nyq}}}{k_0}\right) \approx \arcsin\left(\frac{78.5}{250}\right) \approx 0.32 \text{ rad} \approx 320 \text{ mrad}.$$

This is still well below the singular point at $k_{\perp} = k_0$. The transmission factor $\exp(i\sigma V_{\text{proj}})$ places the same sampling demands in Fresnel and ASM, since it depends only on the projected potential. In practice, this means that if a grid is already adequate for high-quality Fresnel multislice in TEM, it is usually also adequate for angular spectrum multislice. Extra ASM-specific sampling only becomes important in regimes much closer to $k_{\perp} = k_0$, such as extreme near-field or soft-X-ray problems rather than the electron-microscopy cases studied here.

5 Conclusion

We have presented a comparison of three split-step multislice propagators of increasing physical complexity, Fresnel multislice, angular spectrum multislice, and the Wave Propagation Method, benchmarked against a direct forward-ODE solution of the full second-order Klein–Gordon equation in two high-angle settings: the Au [100] beam-amplitude verification of Rother and Scheerschmidt [1], extended here from 25 to 100 unit cells, and a convergent-beam diffraction sample in Au [100] at probe semi-angles up to 100 mrad and specimen thicknesses up to roughly 100 nm. In every case the Klein–Gordon ODE served as a non-perturbative reference that makes no split-step or paraxial approximation, allowing us to attribute the observed discrepancies cleanly to the choice of propagator.

The central result is that replacing the Fresnel kernel with the exact angular spectrum kernel recovers much of what is lost in the paraxial approximation. In the Rother–Scheerschmidt verification, Fresnel multislice tracks the Klein–Gordon reference at the central beam only up to about 10 nm of gold and diverges from the 135 mrad off-axis beam after only ~ 4 nm, in agreement with the original paper from Rother and Scheerschmidt [1]. By contrast, the angular spectrum and Wave Propagation methods both follow the ODE reference closely out to 40 nm, and differ from one another only at the largest thicknesses. The convergent-probe CBED study again confirms this result: across 25, 50 and 100 mrad probe semi-angles, the relative CBED-amplitude error of Fresnel multislice exceeds that of angular spectrum and WPM by roughly an order of magnitude beyond ~ 10 nm, while ASM and WPM more accurately reproduce the ODE ground truth throughout. The WPM is consistently the most accurate of the split-step schemes, holding an average error near 3.5% in the CBED simulation but its accuracy advantage over angular spectrum multislice is small in absolute terms and comes at a substantially higher computational cost.

Taken together, these results indicate that the dominant missing physics in conventional Fresnel multislice for TEM is non-paraxial free-space dispersion, not the local, potential-dependent change of wavelength inside the specimen that the WPM additionally captures. This is consistent with the sampling analysis of Section 4: under typical TEM grids and slice thicknesses, the angular spectrum kernel imposes only a few-percent tighter sampling bound than the Fresnel kernel, and the Nyquist scattering angle is far below the singular point $k_{\perp} = k_0$ where the angular spectrum operator becomes brittle. The implication for electron microscopy in at least the high energy regime we have discussed here is that any existing Fresnel multislice code can be upgraded to wide-angle accuracy by changing a single line in the propagator—replacing $\exp(-ik_{\perp}^2 \Delta z / 2k_0)$ with $\exp(i\Delta z \sqrt{k_0^2 - k_{\perp}^2})$ —without changing the FFT grid, the slice thickness, the transmission step, or the per-slice cost.

Looking forward, the Klein–Gordon ODE and WPM methods still provide an, arbitrary-probe ground-truth reference that we expect to be useful beyond the cases studied here, particularly for benchmarking new propagators and for exploring solutions where the local wavelength inside the medium is no longer approximated by its vacuum value—very thick high- Z specimens at low accelerating voltages, soft-X-ray problems, and near-field optical settings. The differentiable, quadratically-binned WPM implementation also keeps the method tractable on GPU and compatible with automatic

differentiation, which could make it a credible forward model for gradient-based inverse problems such as multislice ptychography in the future.

References

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A Differentiable refractive-index binning for WPM multislice

To make the computation of the wave propagation method tractable and limit the need for $O(N^{2D})$ computations, we adapt the WPM algorithm presented in [4] which uses a refractive index binning strategy to limit the number of unique refractive index values in a slice while maintaining differentiability with respect to the refractive index map $n(x, y)$.

To implement this method, we first discretize the refractive index range $[n_{\min}, n_{\max}]$ of the current slice into N_{bins} discrete reference values, denoted as $\{n_{\text{ref}}^{(k)}\}_{k=1}^{N_{\text{bins}}}$.

In a typical multislice potential, many of the refractive index values will be very close to 1 (vacuum region), while the refractive index values of an atom will concentrate in a narrow range slightly above 1. Since there are likely to be many more refractive index values near the vacuum region, we employ a quadratic distribution for the reference indices rather than using linearly spaced bins in order to allocate more bins near $n = 1$. The reference indices are distributed according to a quadratic function determined by the parameter γ , with $\gamma = 2.0$ for quadratic spacing:

$$n_{\text{ref}}^{(k)} = n_{\min} + (n_{\max} - n_{\min}) \left(\frac{k}{N_{\text{bins}} - 1} \right)^\gamma$$

which allows for denser binning near the vacuum refractive index when $\gamma > 1$.

Next, for a propagation step Δz , we compute a bank of propagated fields $u_k(x, y)$. Each field u_k corresponds to the diffraction of the input field $u_{\text{in}}(x, y)$ assuming a homogeneous medium with refractive index $n_{\text{ref}}^{(k)}$. This is performed efficiently via batch FFT operations using the angular spectrum kernel:

$$u_k(x, y) = \mathcal{F}^{-1} \left(\mathcal{F}(u_{\text{in}}) e^{i\Delta z \sqrt{(k_0 n_{\text{ref}}^{(k)})^2 - k_x^2 - k_y^2}} \right)$$

To evaluate the field at a specific spatial coordinate (x, y) where the local refractive index is $n(x, y)$, we identify the two bounding reference bins such that $n_L \leq n(x, y) < n_R$, where $n_L, n_R \in \{n_{\text{ref}}^{(k)}\}$. We adopt a differentiable interpolation scheme by defining a local interpolation weight

$$w_{\text{raw}} = \frac{n(x, y) - n_L}{n_R - n_L}, \quad w_{\text{raw}} \in [0, 1]$$

To ensure smooth gradients and minimize discretization artifacts, we map this linear weight through a smoothstep activation function $S(w)$, akin to the spatial partitioning technique demonstrated by [5] via a cubic Hermite interpolation function:

$$w = S(w_{\text{raw}}) = 3w_{\text{raw}}^2 - 2w_{\text{raw}}^3$$

While linear interpolation would satisfy the requirement for differentiability, it could result in discontinuities in the gradient at the bin boundaries in a gradient-based optimization scheme.

The final field at the next step $u(x, y, z + \Delta z)$ is computed as the weighted superposition of the fields propagated at the bounding indices:

$$u(x, y) = (1 - w) \cdot u_L(x, y) + w \cdot u_R(x, y)$$

This formulation ensures that the propagation model is fully differentiable with respect to the refractive index map $n(x, y)$, enabling inverse reconstruction of potentials via automatic differentiation, and that only a limited number of FFTs (equal to N_{bins}) are required per propagation step, significantly reducing the computational cost compared to a naive WPM implementation. The accuracy of this binned WPM multislice method depends on the number of bins N_{bins} chosen and will change on a case-by-case basis depending on the refractive index distribution of the sample. One can mitigate this potential source of error by performing a test simulation and doubling the number of bins until no difference is observed in the output wavefunction.

In general, the WPM will need a GPU for efficient computation, and we have found that N_{bins} between 64 and 512 are sufficient for the simulations presented in this work. This is a marked improvement over the naive case, as it enables the WPM to simulate many slices with larger grid sizes (512×512 or 1024×1024 pixels) in a reasonable time frame (seconds to minutes), provided the refractive index binning strategy is capable of correctly capturing the refractive index distribution of the sample.